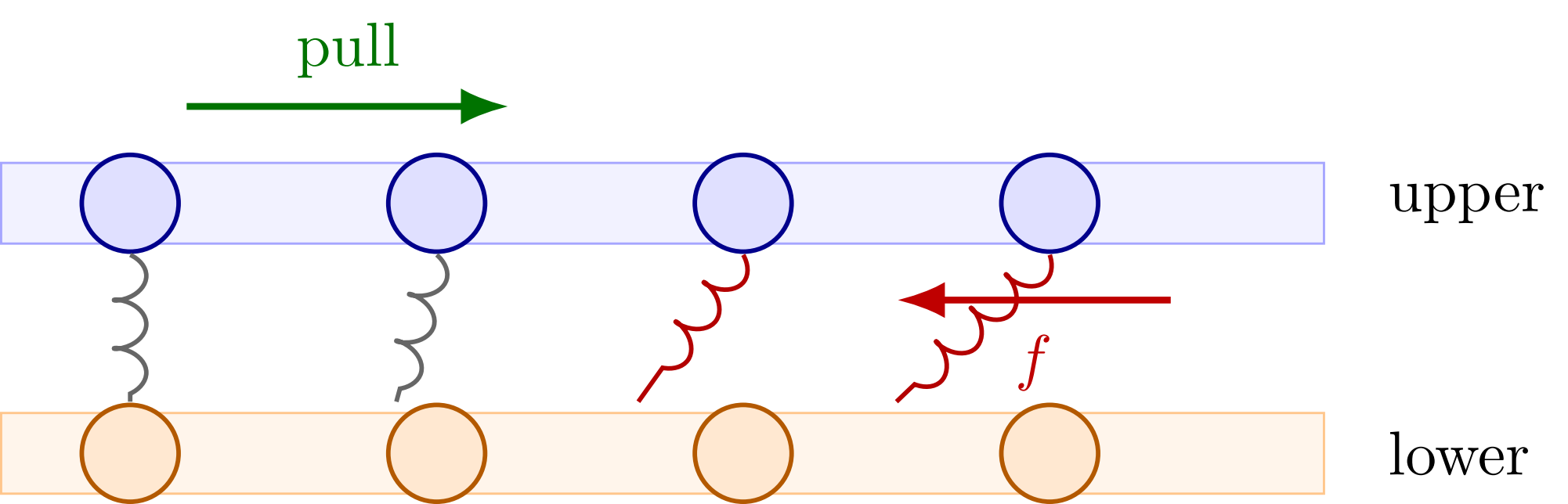
pull
upper
lower
$f$